TASK 10 – COMPARE EXPECTATIONS WITH EXECUTION (4 Marks)

Aim:

To compare the expected and observed results of electricity bill calculations and understand the effects of data types, precision, and overflow.

Matter:

The electricity billing program was tested with different input values to compare the expected results with the actual output.

1. Float and Double: The float data type provides less precision, whereas double provides greater precision for calculations.

2. Integer Overflow: Integer overflow may occur when a calculated value exceeds the maximum limit of its data type, resulting in an incorrect value.

3. Size and Range: The sizes and ranges of int, long, and long long depend on the system and compiler, subject to the C standard requirements.

4. Reasons for Differences: Differences between expected and observed results may occur due to rounding errors, integer overflow, and differences in data-type sizes.

Alternative Concept:

Using double improves floating-point precision. However, storing currency values as integer paise is preferable for accurate billing calculations.

Conclusion:

The program calculates electricity bills and demonstrates rollover consumption using unsigned integer arithmetic. Comparing expected and observed results helps us understand data-type limitations and improve the reliability of billing calculations.

```
#include <stdio.h>

long compute_bill(unsigned int units)
{
    long paise = 5000;

    if (units > 100 && units <= 200)
        paise += (long)(units - 100) * 225;
    else if (units > 200) {
        paise += 100L * 225;
        paise += (long)(units - 200) * 450;
    }

    return paise;
}

int main(void)
{
    unsigned int tests[] = {0, 100, 150, 200, 350};
    int n = sizeof(tests) / sizeof(tests[0]);

    printf("Electricity Billing Test Results\n");

    for (int i = 0; i < n; i++) {
        long bill = compute_bill(tests[i]);
```

```
        printf("%u units -> Rs %ld.%02ld\n",
               tests[i], bill / 100, bill % 100);
    }

    unsigned short previous = 65535;
    unsigned short current = 0;
    unsigned short consumed =
        (unsigned short)(current - previous);

    printf("Rollover consumption: %u units\n",
           (unsigned int)consumed);

    return 0;
}
```

